



Tecnológico de Monterrey

Diseño de sistemas de Control

Actividad:

State Feedback Gain Matrix for Rotatory

Inverted Pendulum

Part A. State Space Modeling

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Stage 1

R.1 According to D.2, derive the linear state space model of the pendulum system.

The chosen state vector was:

$$x = \begin{bmatrix} \theta_r \\ \dot{\theta}_r \\ \theta_p \\ \dot{\theta}_p \end{bmatrix}$$

where:

- θ_r = rotary arm angle (rad)
- $\dot{\theta}_r$ = angular velocity of the rotary arm (rad/s)
- θ_p = pendulum angle measured from upright position (rad)
- $\dot{\theta}_p$ = angular velocity of the pendulum (rad/s)

The input is the motor voltage u , and the outputs are chosen as the arm angle and the pendulum angle:

$$y = \begin{bmatrix} \theta_r \\ \theta_p \end{bmatrix}$$

From the Euler–Lagrange formulation of the Qube Servo 3 pendulum system, the linearized equations of motion around the equilibrium are:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-J_p b_r - \frac{k_m^2}{R_m}}{J_t} x_2 + \frac{m_p^2 l^2 r g}{J_t} x_3 + \frac{m_p l r b_p}{J_t} x_4 + \frac{k_m}{R_m J_t} u \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{m_p l r b_r - \frac{k_m^2}{R_m}}{J_t} x_2 - \frac{m_p g l J_r}{J_t} x_3 - \frac{J_p b_p}{J_t} x_4 + \frac{k_m}{R_m J_t} u\end{aligned}$$

where

- $J_t = J_p + J_r - (m_p l^2 + m_r r^2)$ = total inertia term
- R_m = motor resistance
- k_m = motor back-emf constant
- b_r, b_p = damping coefficients

$$\dot{x} = Ax + Bu$$

This leads to the general state-space representation:

$$y = Cx + Du$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-J_p b_r - \frac{k_m^2}{R_m}}{J_t} & \frac{m_p^2 l^2 r g}{J_t} & \frac{m_p l r b_p}{J_t} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m_p l r b_r - \frac{k_m^2}{R_m}}{J_t} & \frac{-m_p g l J_r}{J_t} & \frac{-J_p b_p}{J_t} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{k_m}{R_m} \\ 0 \\ \frac{k_m}{R_m} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = [0]$$

R.2 Based on D.3, compute the state space representation matrices derived in D.2.

Matlab code for State Space model:

Setting Variables and Parameters

```

14. Rm = 7.5;
15. % Current-torque (N-m/A)
16. kt = 0.0422;
17. % Back-emf constant (V-s/rad)
18. km = 0.0422;
19. %
20. % Rotary Arm
21. % Mass (kg)
22. mr = 0.095;
23. % Total length (m)
24. r = 0.085;
25. % Moment of inertia about pivot (kg-m^2)
26. Jr = 2.2879*10^(-4);
27. % Equivalent Viscous Damping Coefficient (N-m-s/rad)
28. br = 1e-3; % damping tuned heuristically to match QUBE-Sero 2 response
29. %
30. % Pendulum Link
31. % Mass (kg)
32. mp = 0.024;
33. % Total length (m)
34. Lp = 0.129;
35. % Pendulum center of mass (m)
36. l = 0.0645;
37. % Moment of inertia about pivot (kg-m^2)
38. Jp = 1.3313*10^(-4);
39. % Equivalent Viscous Damping Coefficient (N-m-s/rad)
40. bp = 5*10^(-5); % damping tuned heuristically to match QUBE-Sero 2 response
41. % Gravity Constant
42. g = 9.81;
43.
44. % Find Total Inertia
45. Jt = (Jp*Jr)-((mp^2)(l^2)(r^2));

```

Computation of matrixes A, B, C and D

```

%% Step 2: Program and compute the State Space Representation matrixes
A = [0 1 0 0;
     0 0 0 1;
     0 ((-Jp*br)-((km*km)/Rm))/Jt (mp^2)*(l^2)*r*(g/Jt) mp*l*r*(bp/Jt);
     0 ((mp*l*r*br)-((km*km)/Rm))/Jt (-mp*g*l*Jr)/Jt -Jp*bp/Jt];

B = [0;
     km/Rm;
     0;
     km/Rm];

C = [1 0 0 0;
     0 0 1 0];
D = [0];

sys = ss(A,B,C,D)

```

Result for matrixes:

```

A =
      x1      x2      x3      x4
x1      0      1      0      0
x2      0      0      0      1
x3      0     152    -12.53    0.5005
x4      0   -264.3     9.857   -0.5064

B =
      u1
x1      0
x2      0
x3    56.98
x4   -56.32

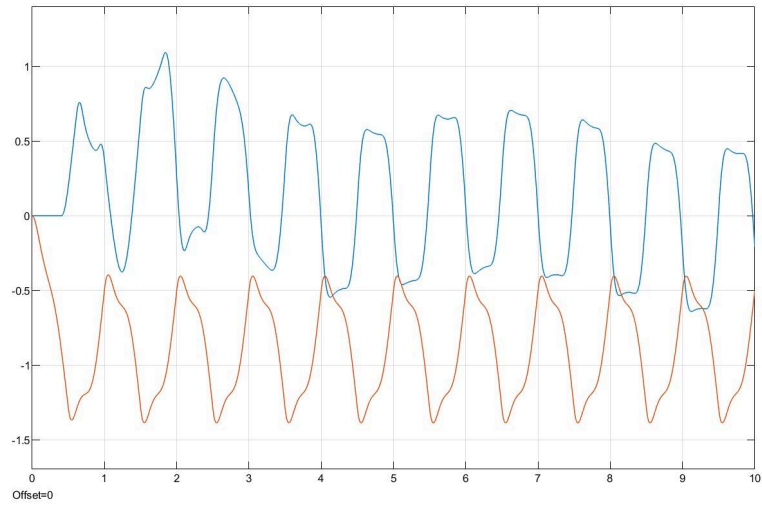
C =
      x1  x2  x3  x4
y1      1   0   0   0
y2      0   1   0   0

D =
      u1
y1      0
y2      0

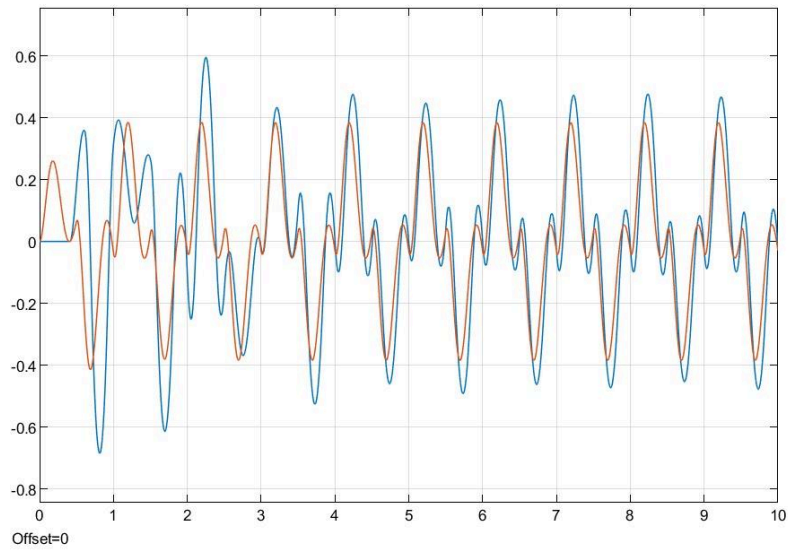
```

R.3 Based on the Virtual Quanser® Qube Servo 3 - Pendulum in D.1 and your state space model, obtain the scopes shown in Fig. 2.3, which represents the response of the rotary pendulum system for the virtual lab and the model and.

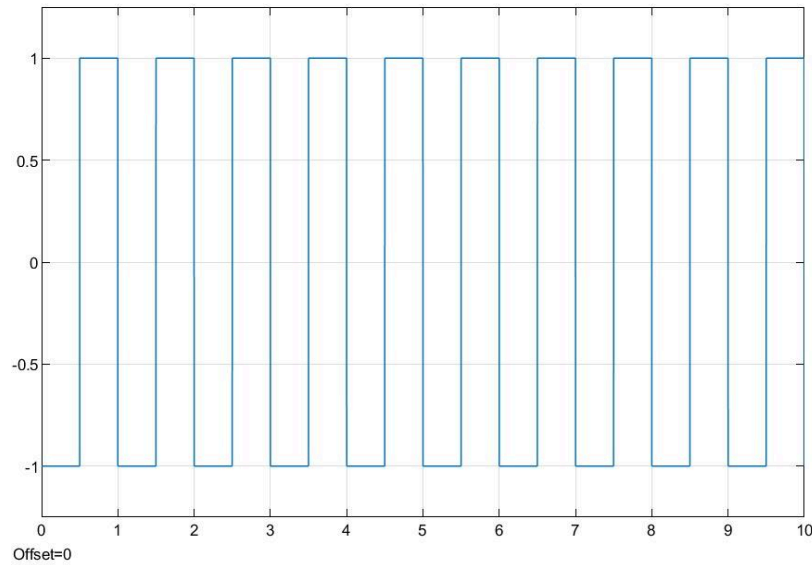
*Note: Add your Matlab® code used to compute these values.



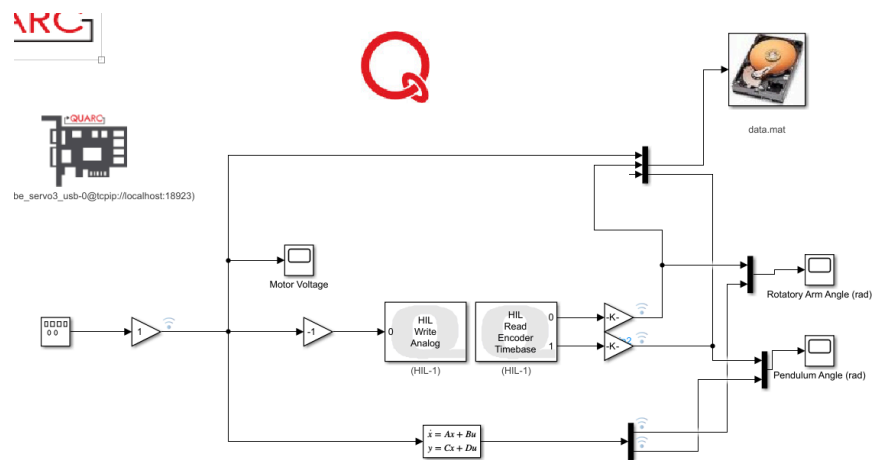
Response of the of the rotatory arm (Simulation and State Equations)



Response of the of the rotatory pendulum (Simulation and State Equations)



Step response of the rotatory pendulum



Configuration on simulink

We can observe that, through state-space modeling, we are able to predict the behavior of the rotatory pendulum with considerable accuracy. However, the values of the rotary arm show some discrepancies, although it is worth noting that the overall trends of the arm are preserved in the prediction.

R.4 Considering D.6, generate plots for the rotary arm response $\theta(t)$ and pendulum response $\alpha(t)$ using the MATLAB script commands and the saved data in the *.MAT files.

R.5 Conclusions.

A01286163 – German Emiliano Rojas Lobatón

This lab really helped me understand state-space models in a more practical way. Before, it all looked like just theory with matrices, but using QUARC and comparing the model with the actual pendulum made everything click. We managed to predict the pendulum angle very well, while the servo angle was harder to capture accurately. Still, I think our results were strong and acceptable.

A01752350 – José Pablo Rodríguez Pinal

What I take away from this activity is how important it is to set up the model correctly and how much that affects the simulation results. There were differences, especially with the servo angle, but the pendulum matched quite closely. For me, the best part was seeing the connection between the equations we derived and the real-life response of the Quanser equipment.

A01830683 – Daniel Santiago Espinosa Cárdenas

I learned a lot about how to interpret each matrix in the state-space model, especially the A and B matrices that define most of the system dynamics. It was interesting to see how these tools allow us to approximate the behavior of such a complex system as the inverted pendulum. The servo prediction wasn't as precise, but the pendulum response was accurate, which shows that the model is definitely useful for control applications.

A01830690 – Daniel Andrés De La Ossa González

With the help of state-space modeling, we obtained a fairly acceptable approximation. We were able to observe that the pendulum angle was predicted accurately (output: theta), but the output alpha could not be predicted with the same level of precision. This suggests that the model captures the main system dynamics, but still requires refinement to improve accuracy in secondary variables.

A01769577 – Monserrath Sánchez Ruiz

I really enjoyed this lab because it was the first time I could actually see how simulations connect with hardware. I noticed it's not easy to make everything match perfectly, but the results

were still good. The pendulum prediction was solid, while the servo was harder to get right, and that made me realize how difficult it is to model every detail. In the end, I think this was a great introduction to how control engineering works in real life

Stage 2

R.6 According to D.2, which are the desired poles based on the required performance index

Specifications?

It is considered to compute the Qube Servo 3 – Pendulum calculations under the following time domain specifications when the pendulum is balanced and tracking a rotary arm setpoint.

$$\%MP = 6.81\%$$

$$t_{ss} = 1.54 \text{ sec}$$

In order to obtain the desired poles, values of σ and W_d must be computed.

$$p_{1,2} = \sigma \pm jW_d$$

Since

$$\%MP = \left\{ e^{-\sigma \frac{\pi}{W_d}} \right\};$$

$$t_{ss} = \frac{4}{\sigma};$$

then

$$1.54 \text{ s} = \frac{4}{\sigma}$$

$$\sigma = \frac{4}{1.54 \text{ s}}$$

$$\sigma = 2.5974$$

$$6.81\% = \left\{ e^{-\sigma \frac{\pi}{W_d}} \right\}$$

$$0.0681 = \left\{ e^{-\sigma \frac{\pi}{W_d}} \right\}$$

$$\ln|0.0681| = -\sigma \frac{\pi}{W_d}$$

$$W_d = -2.5974 \frac{\pi}{\ln|0.0681|}$$

$$W_d = 3.03708$$

Therefore, the desired poles based on the required performance index are:

$$p_{1,2} = -2.5974 \pm j3.03708$$

R.7 Based on D.3, present the state space representation of the Rotary pendulum model in Inverted. Configuration and the corresponding computations needed to verify if the system can be Controlled.

*Note: Add your Matlab® code used.

Computations of Matrixes A, B, C and D when the pendulum is inverted:

```

46. %% Step 2: Program and compute the State Space Representation matrixes
47. A = [
48.     0, 0, 1, 0;
49.     0, 0, 0, 1;
50.     0, (mp^2*l^2*r*g)/Jt, -(Jp*br)/Jt - (Jp*(km^2/Rm))/Jt, -(mp*l*r*bp)/Jt;
51.     0, (mp*g*l*Jr)/Jt, -(mp*l*r*br)/Jt - (mp*r*l*(km^2/Rm))/Jt, -(Jp*bp)/Jt;
52. ];
53.
54.
55. B = [0;
56.     0;
57.     (Jp*km)/(Jt*Rm);
58.     (mp*r*l*km)/(Jt*Rm)];
59.
60. C = [1 0 0 0;
61.     0 1 0 0];

```

62. $D = 0;$

Matrixes results:

A =

```
0         0    1.0000         0
0         0         0    1.0000
0 152.0030 -12.5321   -0.5005
0 264.3013 -12.3862   -0.5064
```

B =

```
0
0
56.9835
56.3201
```

C =

```
1    0    0    0
0    1    0    0
```

D =

```
0
```

It is remarkable that the values of the matrixes are almost the same as the values of the matrixes for the not inverted pendulum (1st Stage), only that this time some of the values of the State and Input Matrixes have inverted signs due to the changes in the equations.

In order to verify if the system can be controlled, the rank of the Controllability Matrix must be equal to the size of the matrix A.

64. %%Matriz de controlabilidad

65. $Q = [B \ A \ B \ (A \cdot A) \cdot B \ (A \cdot A \cdot A) \cdot B];$

66.

67. %%Verifica controlabilidad del sistema

```

68. rango = rank(Q);
69.
70. if rango ~= size(A, 1)
71.     fprintf('El sistema no es controlable. (rango(Q) es diferente de n')
72. elseif rango == size(A, 1)
73.     fprintf('El sistema es controlable. (rango(Q) es igual a n)')

```

Since

$$\text{rank}(Q) = 4 \text{ and } n = 4 ,$$

then

$$\text{rank}(Q) = n$$

Therefore, the system is controllable.

R.8 Based on D.4, enter the state feedback gain matrix computed by the Transformation Matrix T and the Companion Method.

```

El sistema es controlable. (rango(Q) es igual a n)
Matriz de ganancias K Tranformation Matrix T:
-4.4226    49.2709    -1.8678     3.2597

```

```

Matriz de ganancias K con Companion Method:
-4.4226    49.2709    -1.8678     3.2597

```

After applying the companion method, we verified that both methods yielded the same values of k.

```

1.  %% --- Cálculo de la matriz de ganancia K con Companion Method (state feedback) ---
2.
3.  A_Companion = [0 1 0 0;
4.               0 0 1 0;
5.               0 0 0 1;
6.               -a(5) -a(4) -a(3) -a(2)
7.               ];
8.
9.  B_Companion = [0;

```

```

10.         0;
11.         0;
12.         1
13.     ];
14.     det = expand(simplify(det(s*eye(4) - A_Companion)));
15.     polinomio_Companion = sym2poly(det);
16.
17.     W_Companion = [polinomio_Companion(4) polinomio_Companion(3)
polinomio_Companion(2) 1;
18.     polinomio_Companion(3) polinomio_Companion(2) 1 0;
19.     polinomio_Companion(2) 1 0 0;
20.     1 0 0 0];
21.
22.     Q_Companion = [B_Companion A_Companion*B_Companion
(A_Companion*A_Companion)*B_Companion
(A_Companion*A_Companion*A_Companion)*B_Companion];
23.     T_Companion = Q_Companion*W_Companion ;
24.     K_Companion = [alpha(5)-polinomio_Companion(5), alpha(4) -
polinomio_Companion(4), alpha(3) - polinomio_Companion(3), alpha(2) -
polinomio_Companion(2)]*inv(T_Companion);
25.
26.     Last_W = T*inv(T_Companion);
27.
28.     K = double(K_Companion*inv(Last_W));
29.     % Mostrar resultado
30.     fprintf('\nMatriz de ganancias K con Companion Method:\n');
31.     disp(K);
32. end

```

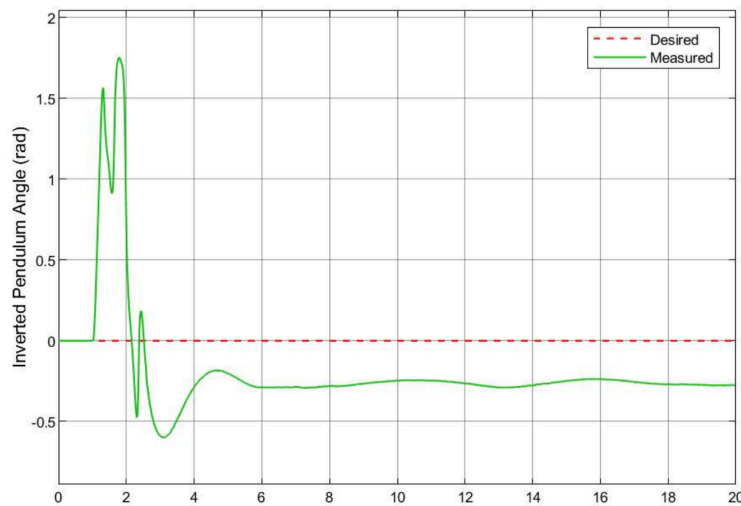
R.9 Considering D.5, generate plots for the controlled rotary arm response $\theta(t)$ and pendulum response $\alpha(t)$ using the MATLAB script commands and the saved data in the *.MAT files.

To validate the performance of the rotary inverted pendulum control system implemented in Simulink and QUARC, the following experimental tests were carried out. Each test allowed us to analyze how the controller behaves under different state feedback gains.

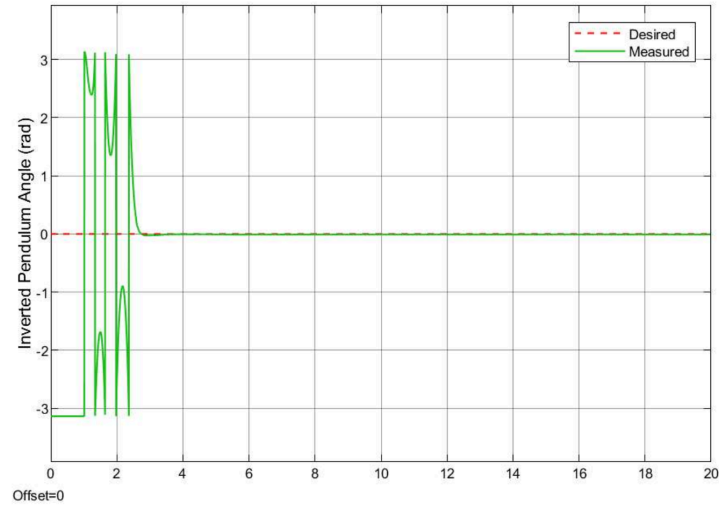
To obtain the data of the simulation we set the gain block to $[0 \ 0 \ 0 \ 0]$, and used the state feedback gain of $K = [-2 \ 50 \ -1 \ 3]$.

Figures below show the measured pendulum angle, rotary arm angle, and motor voltage during the experiment.

At the beginning of the response, high oscillations appear as the pendulum transitions from a non-upright position to the vertical configuration. Once the balancing controller acts, the system rapidly stabilizes and both angles stay close to zero radians.

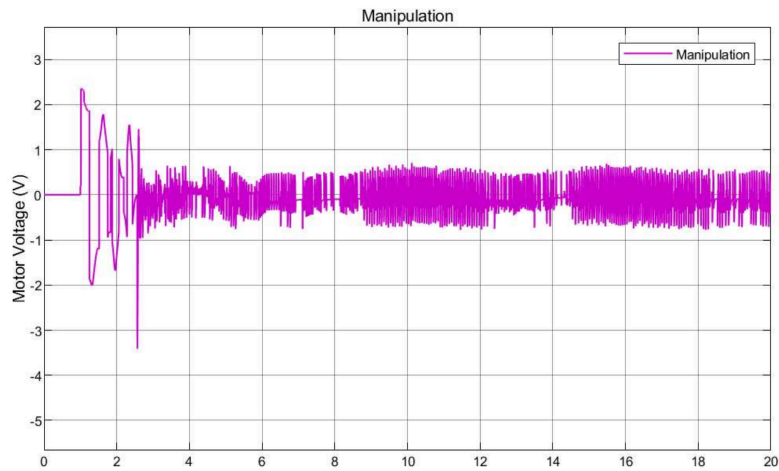


Rotatory arm with given k values



Inverted pendulum with given k values

Regarding the control effort, the motor voltage briefly experiences large variations during stabilization, however, it stays within the actuator limits once the pendulum is balanced. This behavior is expected since the controller must compensate for the initial perturbations to position the pendulum vertically.

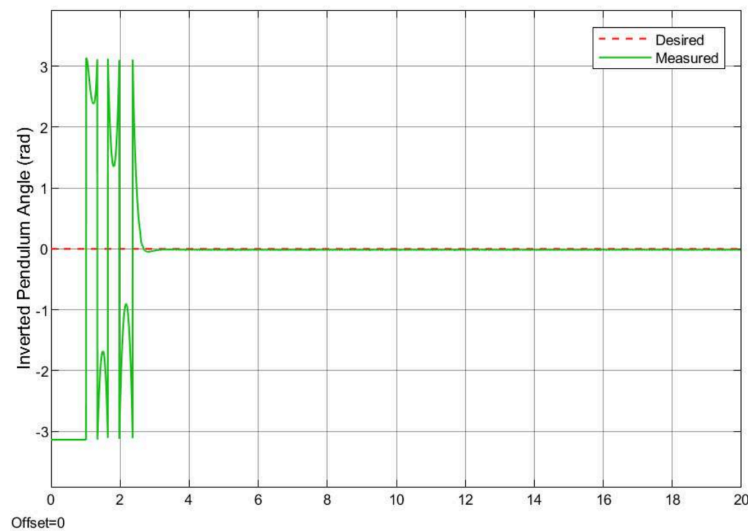


Voltage with given k values

Then we implemented the state-feedback gain that we designed in Section D.3. In this case, $K = [-4.4226 \ 49.2709 \ -1.8678 \ 3.2597]$

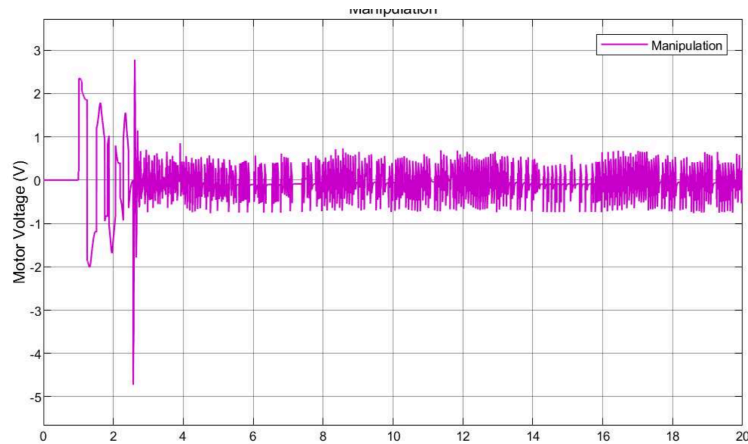
The system behavior can be observed in the plots shown next.

From the plots, we can see that the pendulum starts with some strong oscillations during the first seconds, but then it quickly gets to the vertical position and stays there pretty well. There is almost no steady state error once it stabilizes. The system keeps both the pendulum and the rotary arm under control the whole time.



Inverted pendulum with k values obtained

Something that is very noticeable is the control effort. The motor voltage has higher and faster changes compared to the default gain. This makes sense because our design placed the poles further to the left, so the controller reacts more aggressively to reach equilibrium sooner. Basically, we get a better transient response, but with more work from the motor.



Voltage with k values obtained

In summary, our controller does stabilize the pendulum successfully and it reacts faster than the default one, but it also demands more control action to achieve that.

Matlab code

```

93.  %% --- Cálculo de la matriz de ganancia K (state feedback) ---
94.
95.  del = [alpha(5) - a(5), alpha(4) - a(4), alpha(3) - a(3), alpha(2) - a(2)];
96.
97.  %% Step 4: Matriz de transformación T
98.
99.  W = [a(4) a(3) a(2) 1;
100.      a(3) a(2) 1 0;
101.      a(2) 1 0 0;
102.      1 0 0 0];
103.
104.  T = Q*W ;
105.
106.
107.
108.  % Cálculo de la matriz K
109.  K = del * inv(T);
110.
111.
112.
113. % Mostrar resultado
114. fprintf('\nMatriz de ganancias K:\n');

```

```
115. format short
116. disp(K);
117.
118.
119.
120. end
```

R.10 Personal conclusions.

A01286163 – German Emiliano Rojas Lobatón

Throughout this complete challenge, I was able to clearly see how theoretical control concepts progressively evolve into real implementations. In Stage 1, modeling helped me understand how the pendulum dynamics can be represented mathematically, while Stage 2 showed how those same equations directly influence the controller behavior of the real equipment. The most meaningful part for me was observing the stability achieved through our own gain design. This proved that our work was correct and helped me understand better how state feedback truly allows us to control systems that would otherwise be unstable.

A01752350 – José Pablo Rodríguez Pinal

This project helped me connect modeling, simulation, and practical experimentation. I learned how selecting desired poles affects control performance and how the system responds differently depending on the gain we design. Comparing the default controller against our gain made it clear how the balance between settling time and control effort changes depending on the tuning. Working with the real pendulum made the results much more tangible and helped me better understand the value of validating controllers on hardware. Overall, I feel this activity strengthened my understanding of control design and how each decision directly affects system behavior.

A01830683 – Daniel Santiago Espinosa Cárdenas

During all the stages of this work, I deepened my knowledge of state-space representation and learned to interpret how each matrix affects the dynamics of the pendulum. What I liked the most was seeing the real-time stabilization, because it confirmed that the model is accurate enough for practical control applications. It was also important to observe the limitations in the predictions of certain states, which reminds me that control design always requires refinement. I now understand better why inverted pendulums are such a fundamental benchmark system in control engineering.

A01830690 – Daniel Andrés De La Ossa González

From my perspective, the most valuable part of this challenge was learning how each design step, not just the controller affects the final performance. Obtaining the controllability matrix, defining performance requirements, and then synthesizing the feedback gain made me appreciate the methodology behind control engineering. Testing our gain confirmed the trade-off between faster stabilization and increased voltage demand. This helped me understand that an efficient controller is not only about achieving balance but also about ensuring a reasonable use of actuation resources.

A01769577 – Monserrath Sánchez Ruiz

This project allowed me to see how many engineering areas work together in a single system: sensing, signal conditioning, simulation, and control. Stage 1 helped me understand the importance of accurate modeling when predicting behavior, while Stage 2 showed me how a controller can successfully maintain an unstable system upright. Seeing the pendulum move and then stabilize thanks to our designed gain made me appreciate how powerful feedback control really is. This experience motivates me to keep learning more about real-time control and applying these concepts to more complex systems.

Co-evaluation:

Table 1.1: José Pablo's evaluation of his colleagues

José Pablo Rodríguez Pinal A01752350 Evaluation	Stage 1	Stage 2	Lab Attendance
German Rojas	5	5	5
Daniel Espinosa	5	5	5
Daniel de la Ossa	5	5	5
Montserrat Sánchez	5	5	5

Table 1.2: German's evaluation of his colleagues

German Emiliano Rojas Lobatón A01286163 Evaluation	Stage 1	Stage 2	Lab Attendance
José Pablo Rodríguez	5	5	5
Daniel Espinosa	5	5	5
Daniel de la Ossa	5	5	5
Montserrat Sánchez	5	5	5

Table 1.3: Daniel Sanitago's evaluation of his colleagues

Daniel Santiago Espinosa Cárdenas A01830683 Evaluation	Stage 1	Stage 2	Lab Attendance
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German Rojas	5	5	5
José Pablo Rodríguez	5	5	5
Daniel de la Ossa	5	5	5
Montserrat Sánchez	5	5	5

Table 1.4: Daniel Andrés's evaluation of his colleagues

Daniel Andrés De La Ossa Gonzales A01830690 Evaluation	Stage 1	Stage 2	Lab Attendance
German Rojas	5	5	5
Daniel Espinosa	5	5	5
José Pablo Rodríguez	5	5	5
Montserrat Sánchez	5	5	5

Table 1.5: Montserrath's evaluation of his colleagues

Montserrath Sánchez Ruiz A01769577 Evaluation	Stage 1	Stage 2	Lab Attendance
German Rojas	5	5	5
Daniel Espinosa	5	5	5
José Pablo Rodríguez	5	5	5
Daniel de la Ossa	5	5	5